

Yichao HUANG

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Research interest

Probability theory with applications to mathematical physics, especially on Gaussian Multiplicative Chaos, Liouville Conformal Field Theory and their interactions with other disciplines.

Work and Education

2022 – now	Professor / Research Fellow at Beijing Institute of Technology, China.
2018 – 21	Post-doctoral researcher at University of Helsinki, Finland. Mentors : Antti Kupiainen and Eero Saksman under the ERC-grant QFPROBA.
2014 – 17	PhD student at University Paris VI and DMA of École Normale Supérieure de Paris. Supervisors : Rémi Rhodes and Vincent Vargas . PhD thesis in probability on <i>Gaussian Multiplicative Chaos</i> and <i>Liouville Quantum Gravity</i> .
2010 – 14	Élève normalien at École Normale Supérieure de Paris. FIMFA cursus of École Normale Supérieure de Paris. ENS Paris Diploma. Master “Probabilités et modèles aléatoires” at University Paris VI, grade “Très Bien”.
2008 – 10	Preparatory Classes MPSI-MP*, Lycée Janson de Sailly, Paris. Admitted to École Normale Supérieure de Paris through French national entrance exam.

Publications (last update: April 2026)

- [14] *Royen’s proof of the Gaussian correlation inequality as a supersymmetric dimensional reduction*
arXiv preprint, 2026+
- [13] *The $H^{2|2}$ monotonicity theorem revisited*
with Xiaolin Zeng, **submitted**, 2026+
- [12] *Fine-Graining and Continuous Space Scaling Limit of the $H^{2|2}$ Model on the Hierarchical Lattice*
with Jinglin Wang, Xiaolin Zeng, **submitted**, 2025+
- [11] *Tail profile of bulk Gaussian multiplicative chaos measures in the exact scale-invariant case*
with Youtao Liu, **submitted**, 2025+
- [10] *Left Tail of the Subcritical Derivative Martingale in a Branching Wiener Process*
with Xinxin Chen, Heng Ma, **submitted**, 2025+
- [9] *A moment bound on bulk/boundary quotients of Gaussian multiplicative chaos*
Electronic Journal of Probability, 2026
- [8] *A moment bound on bulk/boundary quotients of Gaussian multiplicative cascades*
with Youtao Liu, **Acta Mathematica Sinica, English Series**, 2026
- [7] *Reinforced loop soup via Wilson’s algorithm*
with Yinshan Chang, Dang-Zheng Liu, Xiaolin Zeng, **Annales Henri Poincaré**, 2026
- [6] *Random analytic functions via Gaussian multiplicative chaos*
with Eero Saksman, **Arkiv för Matematik**, 2025
- [5] *Moment bounds for Gaussian multiplicative chaos with higher-dimensional singularities*
Annales de l’Institut Henri Poincaré (B), 2025
- [4] *Ward identities in the $\mathfrak{sl}_3$ Toda conformal field theory*
with Baptiste Cerclé, **Communications in Mathematical Physics**, 2022
- [3] *Another probabilistic construction of Φ^{2n} in dimension 2*
Electronic Communications in Probability, 2021
- [2] *Two perspectives of the 2D unit area quantum sphere and their equivalence*
with Juhan Aru and Xin Sun, **Communications in Mathematical Physics**, 2017
- [1] *Liouville Quantum Gravity on the unit disk*
with Rémi Rhodes and Vincent Vargas, **Annales de l’Institut Henri Poincaré (B)**, 2018

Supervisions

2024 – now Zhouxing Cai, PhD thesis, on quasiconformal maps and Gaussian multiplicative chaos.

Research visits and mini-courses

- Spring 2025 **Beijing Normal University, China**
Short lecture series on Gaussian Multiplicative Chaos (2 weeks, 12 hours)
- Fall 2024 **Université de Strasbourg, France**
Hosted by Institut de recherche mathématique avancée (IRMA) (2 months)
- Aug 2023 **Sun Yat-sen University, China**
Short lecture series on Gaussian free field (2 weeks, 12 hours)
- Dec 2018 **Tsinghua University, China**
Yau Mathematical Sciences Center (2 weeks)
- Fall 2017 **Institut Mittag-Leffler, Sweden**
IML postdoctoral fellowship grant (3 months)
- Apr 2015 **Isaac Newton Institute, UK**
Cambridge-PSL French embassy fund (2 weeks)

Invited conference talks

- 2027 Mathematical Aspects of Quantum Theory, **TSIMF, Hainan, China**
(Seminars) ..., etc.
- 2026 The 10th International Congress of Chinese Mathematicians (Sessional), **SIMIS, Shanghai, China**
Conformal Field Theories: Randomness and Geometry, **IMS, National University of Singapore**
The 15th American Institute of Mathematical Sciences Conference, **Athens, Greece**
Inaugural Hong Kong Probability Workshop 2026, **The University of Hong Kong**
(Seminars) USTC, Dalian TU, BIT, etc.
- 2025 Beijing-Tianjin-Hebei “JJJ” annual meeting in mathematics, **Nankai, Tianjin, China**
Random Geometry and Topology, **RIMS, Kyoto, Japan**
International Workshop on Nonlinear Stochastic Analysis and Controls, **SJTU, Shanghai, China**
Fujian Young Probabilists Workshop, **Fujian Normal University, China** (cancelled)
(Seminars) USTC, Wuhan, Aalto, AMSS, etc.
- 2024 Workshop on “Probability and Statistical Physics”, **TSIMF, Hainan, China**
Young Forum of Innovation and Technology, **Moscow State University, Russia** (cancelled)
Kuangou Probability Forum, **Kuangou (Beijing), China**
Random Fields and Processes on Graphs and Fractals, **RIMS, Kyoto, Japan**
AMSS conference on Stochastic Analysis and Statistical Physics, **Beijing, China**
The 14th American Institute of Mathematical Sciences Conference, **Abu Dhabi, UAE**
(Seminars) Shanghai, Shenzhen, Great Bay, etc.
- 2023 Second International Conference for Chinese Young Probability Scholars, **Online**
Workshop on “Renormalization and related topics”, **Harbin, China**
Workshop on “SPDEs and related topics”, **Beijing, China**
Masterclass in random geometry by Scott Sheffield, **Beijing, China**
18th International Workshop on Markov Processes and Related Topics, **Tianjin, China**
8th Chinese Conference on Probability Theory, Invited Session, **Fujian, China**
Workshop on “Interacting particle systems, scaling limits and universality”, **Singapore**
- 2022 Conference on probability and related fields in Beijing area, **Beijing, China**
- 2021 First International Conference for Chinese Young Probability Scholars, **Online**
- 2020 Spectral Theory and Probability in Mathematical Physics, **Strasbourg, France**, cancelled
- 2018 Stochastic Processes and their Applications (SPA), Invited Session, **Göteborg, Sweden**
- 2017 Conference “Fractal geometry and dynamics”, **Institut Mittag-Leffler, Sweden**
- 2016 Conference “Les Probabilités de Demain”, **Institut des Hautes Études Scientifiques, France**
Colloquium “Jeunes Probabilistes et Statisticiens”, **École de Physique des Houches, France**
- 2013 Conference “Des probas de la rue d’Ulm”, **École Normale Supérieure de Paris, France**

Conference organization

- 2024 Workshop “Young scholars in probability”, **Zhuhai, China**
- 2023 Workshop “SPDE and related topics”, **Beijing, China**

Teaching

2024 – 25	Lecturer , undergraduate level, Beijing Institute of Technology (112 hours) Topology (honors), Analysis III (honors and reading group)
2023 – 24	Lecturer , undergraduate level, Beijing Institute of Technology (96 hours) Topology (honors), Analysis II
2022 – 23	Lecturer and assistant , undergraduate level, Beijing Institute of Technology (96 hours) Analysis I (assistant), Linear Algebra I (lecturer), Linear Algebra II (reading group)
2021 – 22	Lecturer and assistant , graduate level, University of Helsinki (75 hours) Probability Theory I (hybrid onsite/online version due to Covid-19)
2020 – 21	Lecturer , undergraduate level, University of Helsinki (60 hours) Analysis Ia: Limits and Differentiation (online version due to Covid-19)
2019 – 20	Lecturer and assistant , undergraduate level, University of Helsinki (75 hours) Analysis Ia: Limits and Differentiation (first English version in the University of Helsinki)
2018 – 19	Assistant , undergraduate level, University of Helsinki (15 hours) Vector Analysis I
2014 – 17	Assistant , undergraduate level, University Paris VI (192 hours) Several courses in Maths, Chemistry, Computer Science
2011 – 12	Oral examiner in Preparatory Classes (60 hours) Lycée Janson de Sailly, Paris

Scientific duties

PhD Jury	Mingchang Liu, Tsinghua University (2023) Yanyan Hu, Beijing Institute of Technology (2025) Zuozheng Zhang, Beijing Institute of Technology (2025)
Undergrad thesis	2 on VRJP (2024), 3 on CILT (2025), 4 on SUSY (2026), ...
Referee	Electronic Journal of Probability Journal of Mathematical Analysis and Applications <u>Frontiers of Mathematics</u> Annales de l'Institut Henri-Poincaré (B) <u>Acta Mathematica Sinica, English Series (*2)</u> <u>Journal of Mathematical Physics</u> <u>Annals of Probability</u> <u>Transactions of the AMS</u> <u>Latin American Journal of Probability and Mathematical Statistics</u> <u>Journal of Differential Equations</u>
2013 – 17	Nominated member of the advisory committee of the laboratory , DMA, ENS Paris
2016	Co-Supervision of undergraduate thesis , exchange student from Tsinghua University

Scientific vulgarisation

2024 – now	Monitor of the BIT student club “ Natural Philosophy Club ”
2016 – 17	Jury , TFJM^2 (Tournoi français des jeunes mathématiciennes et mathématiciens) TFJM^2 : team competition by French high school students on open questions
2010 – 11	Tutor of six high school students from ZEP Paris , TalENS association

Languages

Chinese	Native
English	Fluent
French	Fluent
Finnish	Notions